

$$7.2) \phi(p) = 2p' + x^2 p''$$

$$K_p = \{1, x, x^2\}$$

hládame maticu operátora ϕ vzhľadom k báze K_p

$$\phi(1) = 2(1)' + x^2(1)'' = 0$$

$$\phi(x) = 2(x)' + x^2(x)'' = 2$$

$$\phi(x^2) = 2(x^2)' + x^2(x^2)'' = 2 \cdot 2x + x^2(2x)' = 2x^2 + 4x$$

$$[\phi]_{\mathcal{K}}^{\mathcal{K}} = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

hládame charakteristický polynóm a vlastné čísla ϕ

$$\phi - \lambda I_d = \begin{pmatrix} -\lambda & 2 & 0 \\ 0 & -\lambda & 4 \\ 0 & 0 & 2-\lambda \end{pmatrix}$$

$$\det(\phi - \lambda I_d) = -\lambda \cdot \begin{vmatrix} -\lambda & 4 \\ 0 & 2-\lambda \end{vmatrix} = -\lambda(-\lambda(2-\lambda) - 0) = \lambda^2(2-\lambda)$$

vlastné číslo 0 s alg. násobnosťou 2

vlastné číslo 2 s alg. násobnosťou 1

$$\phi - 0I_d = \phi = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix} \sim \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} \lambda_3 = 0 \\ \lambda_2 = 0 \\ \lambda_1 = 2 \end{array}$$

$$M_0 = \ker \phi = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

$$\phi - 2I_d = \begin{pmatrix} -2 & 2 & 0 \\ 0 & -2 & 4 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} -2 & -2 & 0 \\ 0 & 2 & -4 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} \lambda_2 = 2x_3 = 2s \\ \lambda_1 = x_2 = 2s \end{array}$$

$$M_2 = \ker(\phi - 2I_d) = \text{Span} \left\{ \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \right\}$$

$\lambda = 0$ s geo. násobnosťou 1 $\rightarrow$ 1 Jordanov retiazok

$\lambda = 2$ — — — $\rightarrow$ 1 Jordanov retiazok

$\lambda = 0$ s alg. nás. 2 $\rightarrow$ retiazok dĺžky 2

$\lambda = 2$ s alg. nás. 1 $\rightarrow$ retiazok dĺžky 1

$$\begin{array}{ccc} \vec{v}_2 & \xrightarrow{S} & \vec{v}_1 \xrightarrow{S} \vec{0} \\ & & \vec{v}_3 \xrightarrow{S-2Id} \vec{0} \end{array}$$

$$\vec{v}_3 \in \ker(S - 2Id) \rightarrow \vec{v}_3 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

$$\vec{v}_1 \in \ker(S) \rightarrow \vec{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

kladame $\vec{v}_2$ a bazu B

$$\left(\begin{array}{ccc|c} 0 & 2 & 0 & 1 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} 0 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad \begin{array}{l} \lambda_3 = 0 \\ \lambda_2 = 1/2 \\ \lambda_1 = 0 \end{array}$$

$$\vec{v}_3 = \begin{pmatrix} 0 \\ 1/2 \\ 0 \end{pmatrix}$$

$$B = \left\{ 1, \frac{1}{2}x, 2 + 2x + x^2 \right\}$$

$$[J]_B^B = \left(\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \\ \hline 0 & 0 & 2 \end{array} \right)$$